NA 570 Ammonia Retrofit Project Step #1

Due Wednesday, March 18th

1.0 Design Problem and Background:

A ship owner with an existing 175,000 DWT heavy fuel oil (HFO) bulk carrier vessel has decided to convert it to ammonia. This bulk carrier trades grain between Brazil and China (≈12,000nm each way). With most of the vessel's principal characteristics fixed and the desire to limit technical modifications, the owner would like to maintain the existing hull form and design draft. The ship owner would like to minimize total cost of ownership of the vessel after the retrofit and is willing to consider compromises to endurance, cargo deadweight and speed. The ship owner would like to operate the vessel for 10 years after the retrofit. There are also two ammonia fuel tank locations the owner would like to investigate: 1) on deck aft of the deckhouse and 2) in the cargo hold forward of the deckhouse.

The model will include four independent variables:

- E_D: Endurance using ammonia (nm)
- W_C: Cargo deadweight (tons)
- V: Speed (kt)
- Tank type/location (two discrete values: on-deck or in-hold)

As part of this techno-economic assessment, four technical constraints exist (and need to be evaluated):

1. Space Limitations (ammonia fuel tank volume)
2. Longitudinal Strength (bending moment)
3. Hydrostatics/Weights Balance (added weight)
4. Transverse Stability (GM)

In addition, the economic part of the assessment needs to be completed:

5. Additional total cost of ownership (TCO)

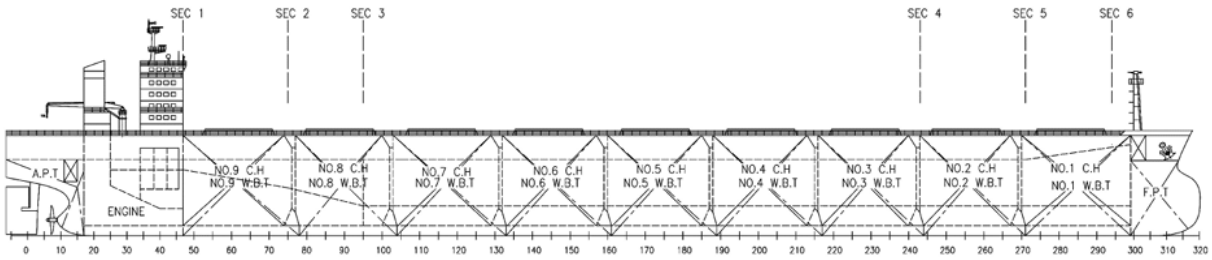
These five outputs from the model will determine the economics and technical feasibility for the independent variable values considered.

2.0 Model Assumptions

2.1 Principal Characteristics

Characteristic	Value	Unit
Length (L_{pp})	278	*m*
Beam (B)	45	*m*
Depth (D)	25.4	*m*
Design Draft (T_D)	17.6	*m*
Block Coefficient (C_B)	0.85	-
Initial Displacement (Δ)	200,000	*t*
Deadweight (DWT)	175,000	*t*
Design Speed (V_D)	14.5	*kt*

2.2 General Arrangement



2.3 Constants

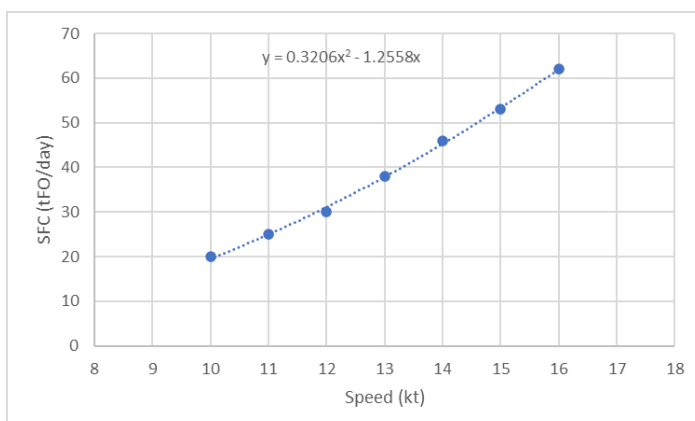
Constant	Value	Unit
Lower Heating Value – Fuel Oil (LHV_{FO})	40.4	GJ/t
Lower Heating Value – Ammonia (LHV_{NH3})	18.8	GJ/t
Density – Fuel Oil (ρ_{FO})	0.93	t/m^3
Density – Ammonia (ρ_{NH3})	0.7	t/m^3

2.4 Specific Fuel Oil Consumption

The owner has provided some actual fuel oil consumption values that we can use for our analysis. For speeds greater than or equal to 10 knots (kt), the following table of actual values and regression equation can be used to estimate the specific fuel oil consumption. Note that speeds below 10 knots should not be explored or considered (too slow for the owner!).

$$SFC_{FO} = 0.3206V^2 - 1.2558V$$

Speed (kt)	SFC (tFO/day)
10	20
11	25
12	30
13	38
14	46
15	53
16	62



3.0 Calculations

3.1 Calculate amount of ammonia fuel needed.

To calculate the amount of ammonia fuel needed (weight and volume), let's start by calculating the equivalent amount of fuel oil needed and convert to the amount of total energy required.

$$Voyage\ Time\ (E_T) = \frac{E_D}{24V} \quad (days)$$

$$FO\ Weight\ (W_{FO}) = SFC_{FO} * E_T\ (tons)$$

$$FO\ Volume\ (\nabla_{FO}) = \frac{W_{FO}}{\rho_{FO}}\ (m^3)$$

$$Required\ Energy\ (E_R) = W_{FO} LHV_{FO}\ (GJ)$$

$$Ammonia\ Weight\ (W_{NH3}) = \frac{E_R}{LHV_{NH3}}\ (tons)$$

$$Ammonia\ Volume\ (\nabla_{NH3}) = \frac{W_{NH3}}{\rho_{NH3}}\ (m^3) \rightarrow \textbf{Output \#1 (Space Limitations)}$$

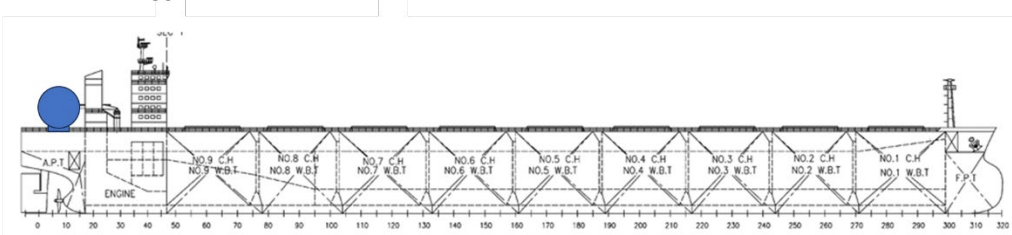
3.2 Calculate the added weights resulting from the retrofit modifications.

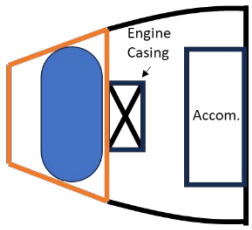
The added weight results from a change in cargo deadweight, the amount of ammonia and steel reinforcement, if needed. It is also assumed that the fuel oil capacity can be half of the original amount (still needed for pilot fuel and redundancy).

The added weight of the ammonia fuel has already been calculated in the previous step, W_{NH3}. We also need to account for the added weight of the ammonia fuel storage tank as well (we assume no other additional weights resulting from ammonia as a fuel are being added).

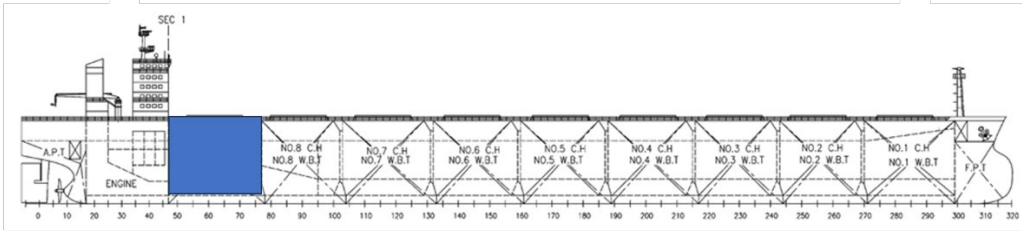
The ammonia storage tank weight can be calculated based on the total volume that needs to be stored (∇_{NH3}) and the type of storage tank. For the on-deck tank location, Type-C (cylindrical) tanks will be used. For the in-hold tank location, Type-A/B (assume rectangular shape) tanks will be used. See below images for more details on the two tank locations and associated tank types.

On-Deck Type C





In-Hold Type A/B

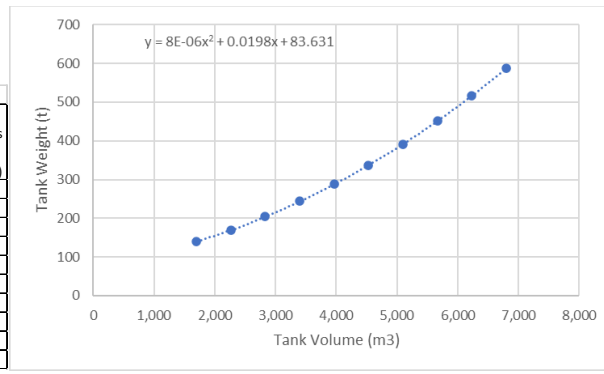


Example #1

Example #2

Below is a table of typical Type-C tank volumes, diameters, lengths and estimated weights. Use this table and associated regression equation to calculate the tank weight for a given volume.

Type-C					
Vessel volume (m3)	Vessel diameter (m)	Length of the tank (m)	Cylindrical shells thickness (mm)	Dished end/Spherical shells thickness (mm)	Total weight of cylindrical shells and dished end of the tank (Ton)
1,699.01	4.267	10.155	9.344	5.176	139.142
2,265.35	4.572	10.881	9.94	5.474	169.78
2,831.68	4.876	11.606	10.535	5.772	204.519
3,398.02	5.181	12.332	11.131	6.07	243.791
3,964.36	5.486	13.057	11.728	6.369	287.756
4,530.70	5.791	13.783	12.324	6.667	336.74
5,097.03	6.096	14.508	12.92	6.966	390.946
5,663.37	6.401	15.233	13.517	7.264	450.671
6,229.71	6.706	15.959	14.113	7.563	516.231
6,796.04	7.008	16.685	14.708	7.86	587.73



$$Tank\ Weight\ (W_{TANK,deck}) = 0.000008\nabla_{NH3}{}^{2} + 0.0198\nabla_{NH3} + 83.631$$

The tank weight for the in-hold location is assumed to be zero as there is existing structure present for the cargo hold.

$$Tank\ Weight\ (W_{TANK,hold}) = 0$$

As mentioned previously, it is assumed that the fuel oil capacity is reduced to half of the original amount. This translates to a negative added weight that should be accounted for.

$$Reduced\ FO\ Weight\ (W_{FO,red}) = -\frac{W_{FO}}{2}$$

For future calculations, the length of the ammonia fuel tank in the cargo hold can be calculated using the height ($H_{hold} = 22$) and depth ($D_{hold} = 38$) of the cargo hold.

$$L_{hold} = \frac{\nabla_{NH3}}{H_{hold}D_{hold}}\ (m)$$

3.3 Calculate the impact of ammonia fuel and storage tank on longitudinal strength

Checking the longitudinal strength of the vessel is important for this type of modification, especially for locations towards the ship's ends. As part of this model, we will assume that additional steel reinforcement is needed depending on the additional moment created by the added weights up to a certain point where additional reinforcement is no longer possible.

With the largest still water bending moment (SWBM) occurring at amidship and in hogging condition, the additional maximum bending moment is to be calculated and compared to the provided SWBM, $M_{SW,H}$. The moments of the changed weights including cargo, ammonia fuel, ammonia fuel tank, and fuel oil can be calculated.

To simplify the calculation for our analysis, we will assume the additional moments can be estimated by modeling the ship as a prismatic beam. The book *Ship Structural Analysis and Design* provides an estimate of the maximum moment change associated with an added weight, W, based on the distance from that added weight to the nearer centroid of the half-waterplane area (d). A sign convention is adopted to account for sagging moments when the load is between the centroid and amidships and hogging moments when it is outside. The moment change, ΔM_{max}, can be estimated as:

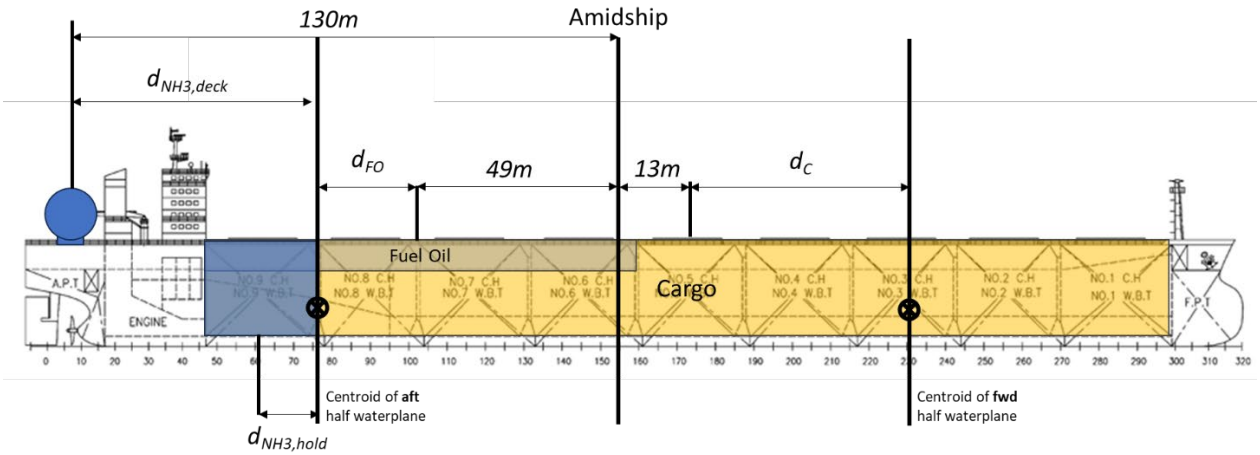
$$\Delta M_{max} \cong \frac{Wd}{2}$$

Additional assumptions:

Characteristic	Value	Unit
Still water bending moment, hogging ($M_{SW,H}$)	4,000,000	*kN-m*
Cargo distance from centroid (d_C)	56.5	*m*
Ammonia (on deck) fuel + tank distance from centroid ($d_{NH3,deck}$)	60.5	*m*
Fuel oil distance from centroid (d_{FO})	20.5	*m*
Density – Grain (ρ_{CARGO})	0.8	t/m^3

The distance from the centroid of the aft half-waterplane area to the ammonia fuel tank for the in-hold option is variable based on the ammonia fuel volume and length of the fuel tank previously calculated:

$$d_{NH3,hold} = 28 - \frac{L_{hold}}{2} \ (m)$$



$$Cargo\ Moment\left(M_{C,deck}\right) = \frac{(W_C - 175{,}000) * d_C}{2}\ (ton\text{-}m)$$

$$Cargo\ Moment\left(M_{C,hold}\right) = \frac{-\nabla_{NH3}\rho_{CARGO} * d_{NH3,hold}}{2}\ (ton\text{-}m)$$

$$Ammonia\ Fuel + Tank\ Moment\ \left(M_{NH3,deck}\right) = \frac{(W_{NH3} + W_{TANK,deck}) * d_{NH3,deck}}{2}\ (ton\text{-}m)$$

$$Ammonia\ Fuel + Tank\ Moment\ \left(M_{NH3,hold}\right) = \frac{(W_{NH3} + W_{TANK,hold}) * d_{NH3,hold}}{2}\ (ton\text{-}m)$$

$$Fuel\ Oil\ Moment\ (M_{FO}) = \frac{-W_{FO,red} * d_{FO}}{2}\ (ton\text{-}m)$$

$$Added\ Moment\ \left(M_{ADD,deck}\right) = M_{C,deck} + M_{NH3,deck} + M_{FO}\ (ton\text{-}m)$$

Because the in-hold option is replacing cargo with ammonia, this needs to be accounted for when calculating the additional weights and moments.

$$Added\ Moment\ \left(M_{ADD,hold}\right) = M_{C,hold} + M_{NH3,hold} + M_{FO}\ (ton\text{-}m)$$

To compare to the provided SWBM in kN-m, convert the added moment from units of ton-m to kN-m by multiplying it by 9.81.

$$Added\ Moment\ (M_{ADD}) = M_{ADD} * 9.81\left(\frac{m}{s^2}\right)\ \ (kN\text{-}m)$$ → **Output #2 (Longitudinal Strength)**

The additional steel weight needed to reinforce the longitudinal structure can be estimated using the below equation:

$$Additional\ Steel\ Weight\ (W_{STEEL}) = 0.036\frac{M_{ADD}}{M_{SW,H}}\Delta\ (tons)$$

3.4 Hydrostatics/weight balance calculation

As part of the weight balance and hydrostatic considerations, the total additional weight is to be calculated.

$$Total\ Additional\ Weight\ (W_{ADD,deck}) = (W_C - 175{,}000) + W_{NH3} + W_{TANK} + W_{FO,red} + W_{STEEL}\ (tons)$$

→ **Output #3 (Hydrostatics/Weight Balance)**

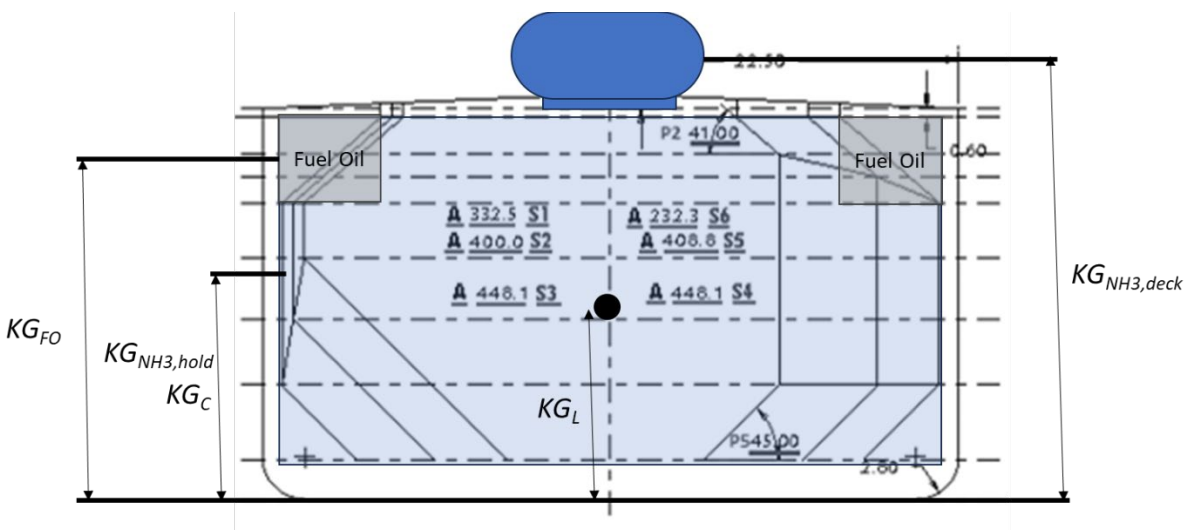
$$Total\ Additional\ Weight\ (W_{ADD,hold}) = -\nabla_{NH3}\rho_{CARGO} + W_{NH3} + W_{TANK} + W_{FO,red} + W_{STEEL}\ (tons) \rightarrow$$

Output #3 (Hydrostatics/Weight Balance)

3.5 Transverse stability calculation

To check for sufficient transverse stability, a new vertical center of gravity (VCG) is to be calculated based on the updated weights already calculated above. Additional assumptions needed for the calculation are provided below:

Characteristic	Value	Unit
Lightship Weight (W_L)	23,645	*t*
Lightship Vertical Center of Gravity (KG_L)	12	*m*
Cargo Vertical Center of Gravity (KG_C)	13	*m*
Fuel Oil Vertical Center of Gravity (KG_{FO})	21.9	*m*
On-Deck Ammonia Fuel/Tank Fuel Vertical Center of Gravity ($KG_{NH3,deck}$)	28.4	*m*
In-Hold Ammonia Fuel/Tank Fuel Vertical Center of Gravity ($KG_{NH3,hold}$)	13	*m*
Metacentric Height (KM)	16	*m*



Vertical moments are calculated and used to determine an updated VCG that can then be used in combination with the metacentric height (KM) from hydrostatic properties of the vessel to calculate the positive metacentric height value (GM).

$$Updated\ Loaded\ Displacement\ (\Delta_{deck}) = W_L + W_C + W_{NH3} + W_{TANK,deck} - W_{FO,red} + W_{STEEL}\ (ton)$$

$$Updated\ Loaded\ Displacement\ (\Delta_{hold}) = W_L + (175{,}000 - \nabla_{NH3}\rho_{CARGO}) + W_{NH3} + W_{TANK,hold} - W_{FO,red} + W_{STEEL}\ (ton)$$

$$Total\ Vertical\ Moment\ (M_{ver,deck}) = KG_L W_L + KG_C W_C + KG_{NH3,deck} W_{NH3} + KG_{NH3,deck} W_{TANK,deck} - KG_{FO} W_{FO,red} + KG_L W_{STEEL}\ (ton\text{-}m)$$

$$Total\ Vertical\ Moment\ (M_{ver,hold}) = KG_L W_L + KG_C(175{,}000 - \nabla_{NH3}\rho_{CARGO}) + KG_{NH3,hold}W_{NH3} + KG_{NH3,hold}W_{TANK,hold} - W_{FO,red} + KG_L W_{STEEL}\ (ton\text{-}m)$$

$$VCG_{deck} = \frac{M_{ver,deck}}{\Delta_{deck}}\ (m)$$

$$VCG_{hold} = \frac{M_{ver,hold}}{\Delta_{hold}}\ (m)$$

$GM = KM - VCG\ (m) \rightarrow$ **Output #4 (Transverse Stability)**

3.6 Calculate the total cost of ownership of the ammonia retrofit project

The TCO is considered on a roundtrip basis. This means that the costs are being calculated for a 24,000nm roundtrip journey. If desired, it could also be calculated on a yearly basis or any period, for example, the expected remaining lifetime of the vessel (e.g., 5 or 10 years).

There are three main components of the total cost of ownership after conversion including the additional capital expenditures (CAPEX), cost of ammonia fuel and potential lost revenue from changes in endurance, speed, or cargo deadweight. The total cost of ownership can be calculated using:

$$C_{TCO} = C_{FUEL} + C_{CAPEX} + C_{REV}$$

Some additional assumptions and needed inputs include:

Characteristic	Value	Unit
Transport Work Revenue (R_{TW})	0.000328	*$/ton-nm*
Blue Ammonia Fuel Price (P_{NH3})	25	*$/GJ*
Newbuild Price (P_{NEW})	65	*$M*
In Hold Tank Price ($P_{TANK,IN\ HOLD}$)	1500	*$/m³*
On Deck Tank Price ($P_{TANK,ON\ DECK}$)	1600	*$/m³*

When considering endurance, we need to ensure the vessel can still deliver on a roundtrip operation (24,000nm). If the endurance using ammonia is below this, then there is a resulting penalty associated with the need to refuel multiple times.

$$E_{RATIO} = \frac{24{,}000}{E_D}$$

$$\#\ of\ Bunkers\ (B_\#) = \lceil E_{RATIO} \rceil\ i.e.,\ round\ up\ to\ nearest\ integer.$$

$$C_{FUEL,TOT} = E_R P_{NH3} E_{RATIO}$$

$$C_{FUEL,PEN} = 0.18{B_\#}^2 - 0.25B_\# + 1.06$$

$$C_{FUEL} = C_{FUEL,TOT} C_{FUEL,PEN}$$

The additional CAPEX of the retrofit will differ based on the tank size and location selected. It is assumed that the other systems onboard that require additional CAPEX are the same. A simplified equation can be used to estimate the additional CAPEX associated with the roundtrip journey used as a basis for the TCO calculation.

$$C_{CAPEX,BASE} = 0.15P_{NEW} * 1000000$$

$$C_{CAPEX,TANK,ON\ DECK} = P_{TANK,ON\ DECK}\nabla_{NH3}$$

$$C_{CAPEX,TANK,IN\ HOLD} = P_{TANK,IN\ HOLD}\nabla_{NH3}$$

$$C_{CAPEX} = \frac{C_{CAPEX,BASE} + C_{CAPEX,TANK}}{(3650/E_T)E_{RATIO}}$$

With the potential for reduced cargo weight and volume (for the "in hold" fuel storage tank location), lost revenue also needs to be calculated and considered as an additional cost. Note that because the density of the cargo (grain) is higher than ammonia, the in-hold option leads to increased cargo reduction due to the volume of the ammonia tank.

$$Cost\ of\ lost\ revenue\ (C_{LR,deck}) = 24(175{,}000 - W_C)VR_{TW}\ \textit{(\$/day)}$$

$$Cost\ of\ lost\ revenue\ (C_{LR,hold}) = 24\nabla_{NH3}\rho_{CARGO}VR_{TW}\ \textit{(\$/day)}$$

$$C_{REV} = E_T E_{RATIO} C_{LR}$$

$$C_{TCO} = C_{FUEL} + C_{CAPEX} + C_{REV} \rightarrow \textbf{Output \#5 (Additional TCO)}$$

4.0 Step #1 Instructions

Perform an initial design exploration of the ammonia retrofit model shown and described above. The goal of design space exploration is to vary the four variables while studying the five outputs. Since the tank type/location variable can take only two discrete values (on-deck or in-hold), consider two separate three-variable problems. For one of the three-variable problems (on-deck or in-hold), perform a 4-level full factorial sampling. For the other, perform a CCD sampling. You may determine a logical range to vary each parameter, and you may choose the form of CCD to use (inscribed etc.). You may also choose to eliminate a point or two from the CCD model (or not put it exactly where CCD says to) for physical or model limitations.

For *each* response (model outputs), start by fitting a simple polynomial model of the form:

$$R = \beta_0 + \beta_1 E + \beta_2 W_C + \beta_3 V$$

Using your choice of the adjusted R^2, coefficient magnitude, or statistical test, try to improve on this simple model by adding either quadratic terms (e.g., $\beta_n V^2$) or cross terms (e.g., $\beta_n EV$) to come up with a compact model for each response. You should have ten models when you are finished (five for the on-deck arrangement and five for the in-hold arrangement). Note that the structure of the models will be similar for both arrangements with key differences associated with the physical arrangement of the ammonia fuel tanks.

For *each* response, provide:

- The points used for your design exploration (non-dimensional and dimensional) and the function values at those points.
- The coefficients of the simple model and form and coefficients of your final model
- The method (R^2, coefficient magnitude, or statistical test) and supporting calculations used to improve the simple model

In a short memo (less than two pages), address the following:

- Describe what the models tell you about the importance of the variables and their interactions. Describe the differences in variable importance and interactions between the two arrangements (on-deck and in-hold), including implications of choosing full factorial versus CCD.
- Consider the reduction of the problem. Do you think the model was oversimplified or something has been missed that could be significant? Are all the variables as defined needed?
- The first four outputs can be considered technical constraints (space limitations, longitudinal strength, hydrostatics/weights balance, transverse stability); however, no explicit constraints have been defined yet. What would you propose as constraints (including specific values, if applicable) to ensure the retrofit will be technically feasible?

In setting up this as a group project, I suggest you coordinate on developing a script in MATLAB, Python, or a similar language so you can quickly evaluate the sampled points (you may need to change your CCD design a few times). This work can be divided into parts, with some group members coding and others doing hand calculations to check the code (always check the code!). Once you have the required data, the group can split up the work on the model fitting step and check each other's results. For submission, a short memo with the requested write up and your code/calculations is all that is needed.